Ajay Venkatesh

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Experience

Amazon **May 2025 – Aug 2025**

Software Development Engineer Intern *Seattle, US*

- Designed a secure UI Console architecture using **Midway SSO, OAuth, and HELIS** to enforce least-privilege, role-based access; presented the solution to the team and received design approval.
- Built serverless infrastructure using **AWS CDK (IaC)** to deploy **REST API Gateway**, Lambda (Java 21), DynamoDB, and S3 with cross-account access; implemented a **CI/CD pipeline** enabling multi-stage (dev/stage/prod) and multi-region deployments with automated promotion and rollback controls.
- Developed the frontend using **React 18 and TypeScript**, integrating AWS UI components, Midway AuthN, and HELIS AuthZ with role-aware views and filtering; delivered a UI that was **40% faster** and significantly more secure than older version.
- Implemented backend APIs using **Java Coral, Smithy models, Dagger DI**, and CBOR serialization; improved response time by **35%** and ensured reliability through comprehensive **unit and integration testing**.

New York University **Aug 2024 – May 2025**

Software Developer (On-Campus) *New York, US*

- Contributed to an AI-driven Remote Patient Monitoring System, awarded **$100K** by the National Science Foundation, USA, employing **React Native, JavaScript**, and Cognito authentication for real-time health monitoring.
- Developed an AI model that identifies food items through mobile images and predicts calorie, protein, and fat intake using **YOLO + OCR** based classification; integrated the model into the app workflow.
- Built parallelized **ETL pipelines** in **Python** to transform wearable NoSQL data into structured SQL, reducing processing time by **60%**, and visualized health trends using **AWS QuickSight** dashboards.

PricewaterhouseCoopers (PwC) **Aug 2021 – Aug 2024**

Senior Associate

- Designed and implemented a cloud-native distributed backend in **C#/.NET** on **Microsoft Azure** for a health platform, integrating Microsoft Dynamics via **REST services** for secure data flow and leveraging event-driven Azure services for scalability and fault tolerance.
- Led delivery across requirements gathering, **design review meetings**, development, testing, and deployment, coordinating with architects, PMs, and business stakeholders to align enterprise solutions with PwC SDLC and quality standards.
- Modernized a legacy onboarding system by building a scalable **B2B platform** with modular services backed by **SQL Server**, migrating core processes to the cloud; integrated data pipelines, email event tracking, and BI reporting, improving automation by **70%**.
- Optimized data pipeline by parallelizing **Excel file imports (100K+ rows)**, cutting processing time from **5 hours to 1** and boosting efficiency by **80%**.
- Executed test workflows on backend modules and recorded findings, contributing to **design reviews** to align solutions with evolving client requirements.

Projects

Flight Delay Prediction | ***PySpark, HDFS, Random Forest, Gradient Boosted Tree***

- Built a scalable **PySpark ML pipeline** on HDFS to process and engineer features from over **7M flight records** using SQL-based aggregation and anomaly detection.
- Performed extensive EDA to identify delay patterns; trained and optimized **Random Forest and Gradient Boosted Tree** models achieving **85%+** accuracy.
- Converted model outputs into actionable insights, enabling data-driven scheduling and operational planning for delay reduction.

Skills

Programming Languages: C#, .NET, Python, Java, JavaScript, TypeScript, SQL, C++
Web & APIs: REST services, React, Node.js
Database: SQL Server, RDBMS, NoSQL, Dataverse
Cloud: Azure (Function Apps, Logic Apps, Storage Blobs, AI Services, Cognitive Services), AWS (CDK, Lambda, S3, EC2, IAM)
Tools: Git, Docker, Visual Studio, Postman, Power BI, CI/CD Pipelines
Methodology: Agile, Scrum, SDLC, Test-Driven Development (TDD), Unit & Integration Testing

Education

New York University | *Master of Science in Computer Engineering* ***(GPA – 3.9/4)*** Aug 2024 – Aug 2026

Certifications

Azure Data Fundamentals
Power Platform Developer Associate
Power BI Data Analyst Associate
Power Platform Solution Architect Expert
Azure Data Scientist Associate
AWS Cloud Practitioner

Publications

Supervised ML System Based Segmentation and Classification of Stroke Using Deep Learning Techniques *(IEEE)*
An Aquila-optimized SVM classifier for Diabetes prediction *(IEEE)*